LOCATION:

- The data pipeline in src/icu_forecast/
- All numbers come from data/processed/cleaning_report.json, data/processed/features_report.json, and the training split
- The file for input to ML model is data/processed, which is the same as the csv file for reading in data/processed_csv (which is selected from the original dataset downloaded from MIMIC).

SELECTION:

From the M2 proposal: for **frequently admitted inpatients**, predict an **unplanned ward-to-ICU transferwithin 24 h and 48 h**, scored once a day, and flag the top 10% highest-risk patients.

This defines what the data must provide: **Need**	**Why**
Admission history per patient	to define "frequently admitted"
Exact location and time of every ward and ICU stay	to build daily ward snapshots and the label
Admission type	to exclude planned (elective) ICU admissions
Clinical signals recorded **on the ward, before** the transfer	to make features that are known at prediction time

I selected 9 files from the original download (hosp/admissions.csv.gz, hosp/patients.csv.gz, hosp/transfers.csv.gz, icu/icustays.csv.gz, hosp/services.csv.gz, hosp/labevents.csv.gz, hosp/prescriptions.csv.gz, hosp/diagnoses_icd.csv.gz, and hosp/omr.csv.gz) and processed them into 10 training files.

#	Processed CSV	Comes from (original download)	Where I covered it
1	admissions_all.csv	admissions	table row "admissions"
2	cohort.csv	admissions + patients + transfers	table row "admissions"
3	transfers.csv	transfers	table row "transfers"
4	labs.csv	labevents (+ d_labitems for names)	table row "labevents"
5	prescriptions.csv	prescriptions	table row "prescriptions"
6	diagnoses.csv	diagnoses_icd	table row "diagnoses_icd"
7	omr.csv	omr	table row "omr"
8	services.csv	services	table row "services"
9	features.csv	calculated from files 2–8	section 5, "new files"
10	split.csv	calculated from cohort	section 5, "new files"

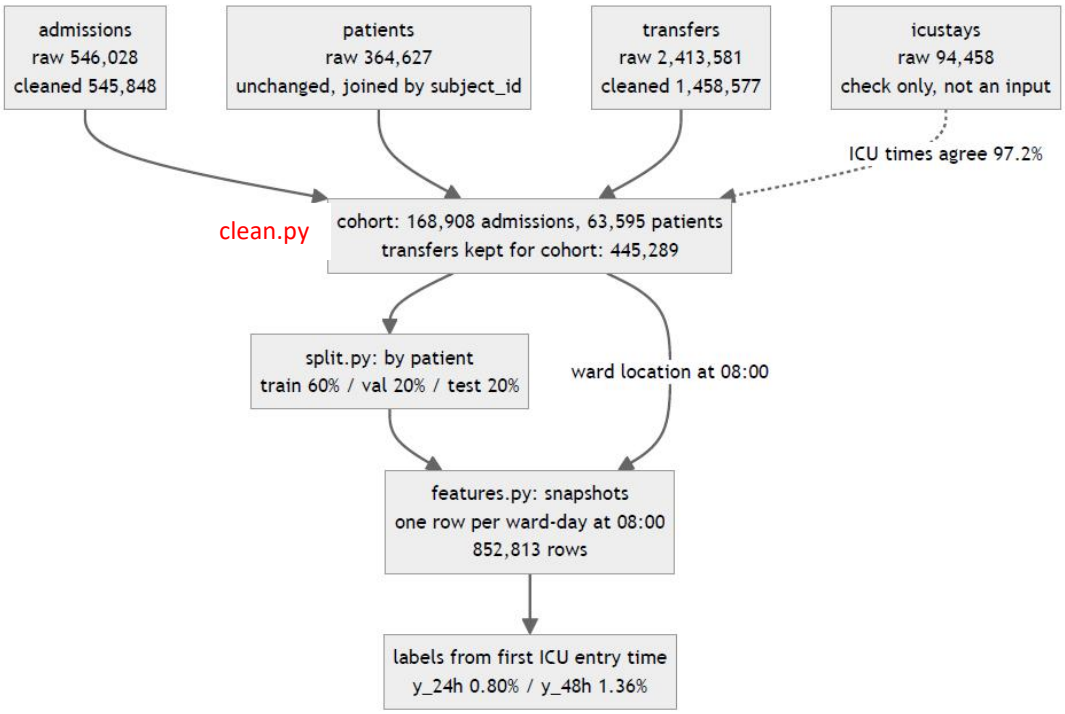
What we kept from the original data:

Table	Original rows	Processed rows	What was kept
admissions	546,028	545,848 (admissions_all) and 168,908 (cohort)	all valid admissions for history; the cohort is readmissions within 365 days, unplanned, with a ward stay, not straight to the ICU
transfers	2,413,581	445,289	only cohort admissions; no discharge rows or rows without an admission
labevents	158,374,764	12,722,653	only cohort admissions and 21 lab tests

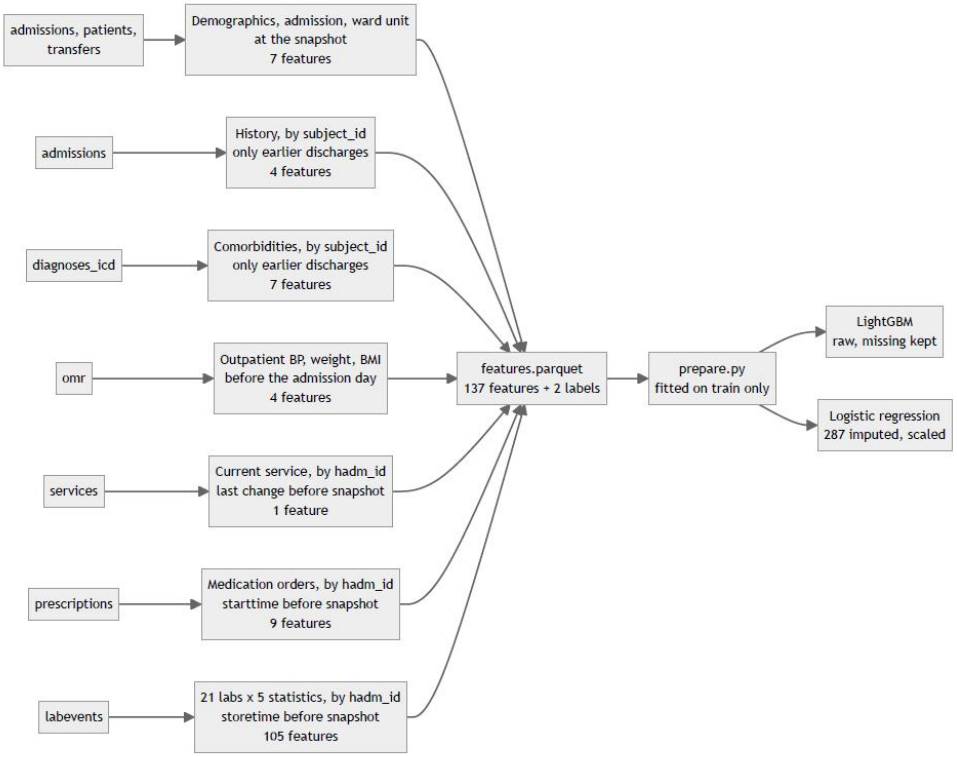
prescriptions	20,292,611	6,273,124	only cohort admissions
diagnoses_icd	6,364,488	4,304,433	only cohort patients
omr	7,753,027	4,729,798	only cohort patients; BP, weight and BMI
services	593,071	184,113	only cohort admissions

The 9 files are processed through clean.py, split.py, and features.py to generate Parquet files for training. These files are saved in data/processed and used as input for the LightGBM or Logistic Regression models.

#	File	Used in
1	hosp/admissions.csv.gz	clean.py: cohort, history
2	hosp/patients.csv.gz	clean.py: sex, age
3	hosp/transfers.csv.gz	clean.py: ward and ICU times, snapshots, labels
4	icu/icustays.csv.gz	clean.py: cross-check of ICU times only
5	hosp/services.csv.gz	clean.py: current service
6	hosp/labevents.csv.gz	clean.py: 21 lab tests
7	hosp/prescriptions.csv.gz	clean.py: medication classes
8	hosp/diagnoses_icd.csv.gz	clean.py: comorbidities
9	hosp/omr.csv.gz	clean.py: outpatient BP, weight, BMI



How are the Parquet files generated after passing the 9 files through clean.py and split.py ?



Meaning of the Features and Why They Were Chosen

Group	**Features**	**Source and join**	**Why it predicts ICUtransfer**	**Known atprediction timebecause**
Demographics, admission	age, sex, admission type andlocation, ward unit, hourssince admission, days sinceprevious discharge	admissions,patients,transfers	older patients and someunits (cardiology, transplant)carry more risk; risk changesover the stay (Figure 6)	recorded atadmission or atthe snapshot
History	prior admissions (total, last365 days), prior ICU stay,length of previous stay	admissions bysubject_id	frequent and recentadmissions mark fragilepatients	only admissionsdischarged before this one
Comorbidities	heart failure, COPD, chronickidney disease, diabetes,cancer, liver disease, sepsis	diagnoses_icdby subject_id	chronic disease raises thechance of decompensation	only diagnosesfrom **earlier**discharges
Outpatient	systolic and diastolic BP,weight, BMI	omr bysubject_id	baseline health status	measured beforethe admissionday
Service	current service	services byhadm_id and time	medical and surgicalservices differ in risk	last servicechange beforethe snapshot
Medications	new orders, new IV orders,and antibiotics, IVantibiotics, vasopressors,diuretics, opioids, insulin,anticoagulants in the last 24h	prescriptionsby hadm_id	escalation of treatment (IVantibiotics, diuretics) signalsdeterioration	starttimebefore thesnapshot
Labs	last value, change sinceprevious, min, max, andcount over 24 h for 21 tests	labevents byhadm_id	abnormal and worseninglabs (lactate, BUN, WBC)and labs being **ordered atall** signal concern	result **released**(storetime) before thesnapshot